

Mahdi Valadan Software Engineer

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Profile

Software engineer who designs, builds and runs complete products. My work covers the whole stack: elegant, carefully crafted interfaces, backend services and APIs, and the servers and infrastructure underneath. I have built geospatial and data-visualization platforms from initial design through deployment and long-term maintenance, and I'm happiest when a project needs both technical depth and visual care. Outside of work, I build games and creative tech projects.

Core Expertise: Full-Stack Product Development & Deployment | Data Visualization | UI/UX Design | Creative Technology

Professional Experience

NHAZCA

Full-Stack Developer | Contract 04/2025 – 06/2026
Rome, Italy

- Designed, built and deployed the web platform for ISABHEL, an ESA-funded structural health monitoring project, integrating satellite InSAR data and ground-sensor measurements for bridges in Turin
- Built interactive map and time-series visualizations that let engineers analyse bridge deformation and structural behaviour over time and across sensors
- Developed the geospatial tile-serving layer with TiTiler, TileServer GL and MinIO, delivering raster and vector tiles to a Mapbox frontend
- Designed and maintained the spatial database on PostgreSQL/PostGIS (Supabase) for satellite and sensor datasets
- Set up and operated production infrastructure on Ubuntu servers using Docker, Coolify and Dokploy, with Vercel for frontend hosting; built backend services and APIs, including transactional email (Mailgun)

Full-Stack Developer | Internship 12/2024 – 03/2025
Rome, Italy

- Development of a data visualization web platform. Integrating WebGIS functionalities and enhancing the platform's capabilities for handling and visualizing geospatial data, and time-series datasets, and delivering an efficient and user-friendly experience.

Front-End Developer | Part-time 10/2022 – 03/2023
Hooshman Institute Tehran, Iran

- Built the front end for an AI-based medical image segmentation tool, used by clinicians and researchers to upload scans and review segmentation results
- Developed responsive, minimal modern UI
- Technology: Tailwind CSS, JS, and Django Template

Full-stack Developer | Internship 07/2021 – 09/2021
Iran National Library Tehran, Iran

- Full-stack Development of a website that showcases the Iran National Library and the ability to download books.
- Technologies: Django, JavaScript, HTML, CSS

Python Developer | Summer Job 06/2020 – 08/2020
K. N. Toosi University of Technology Tehran, Iran

- Contribution to a project titled "Automatic updating of spatial databases using satellite images"
- Technology: Python, QT, and QGIS Library

Education

Master of Computer Science and Engineering 09/2022 – 12/2025
Politecnico di Milano Milan, Italy
- Thesis Title: Modeling Value Detection in Dyadic Conversations for Human-Human and Human-Agent Well-being Coaching
- Selected Courses: Video Game Development and Design, Advanced User Interface, Hypermedia Applications, Artificial Neural Networks and Deep Learning, Design and Implementation of Mobile Applications, and Multidisciplinary Projects.

Bachelor of Computer Engineering 09/2017 – 07/2022
K. N. Toosi University of Technology Tehran, Iran
- Thesis Title: Design and Development of an Interest-free Loan System with Software as a Service Approach
- Teaching Assistant: Internet Engineering (02/2021 – 09/2021), Compiler Design (02/2020 – 07/2020)

Skills

Frontend — JavaScript | TypeScript | React | Vue | Next.js | Nuxt.js | HTML, CSS, Tailwind

Backend — Python | Express.js | Django

Databases — PostgreSQL | PostGIS | Supabase | Firebase | MongoDB

Mobile / Game — Flutter | Unity (C#)

Tools & Platforms — Git | Ubuntu | SSH | Docker | Dokploy | Coolify | Vercel | Mapbox | TiTiler | Tileserver GL | MinIO | QGIS | Mailgun

AI Tools — Codex | Claude | Gemini

Languages

English (Fluent) **Persian** (Native) **Italian** (Basic)

University Projects

Flutter Journal 01/2024 – 07/2024
Social media journaling app
- Politecnico di Milano
- Designer and Developer
- Technology: Flutter, Dart, and Firebase APIs.

The Doomed Dog 10/2023 – 02/2024
3D Action & Adventure PC Video Game
- Politecnico di Milano
- Worked on game mechanics and user interface.
- Technology: Unity Game Engine, and C#

Nico 10/2023 – 02/2024
Interactive Social Learning Game for Children with social disorder and autism
- Politecnico di Milano
- Team Leader
- Developer
- Technology: Unity Game Engine and OpenAI GPT